

两样本独立t检验

假设两组样本 $y_{11}, y_{12}, y_{13}, \dots$

$y_{21}, y_{22}, y_{23}, \dots$

每一组为独立同分布正态分布的随机变量

$$y_{ij} \sim N(\mu_i, \sigma^2)$$

组内之间同期望，组与组之间期望不一定相同，但方差一定相同

独立t检验主要用于检验两样本均值是否一样

t检验小样本的均值与总体均值差异

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

(t值计算)

$\bar{X}$: 样本均值, μ : 总体均值

s : 样本标准差 n : 样本量

自由度 $df = n - 1$

样本总是与自由度相关(无偏估计)

两样本的合并方差: $S_w^2 = \frac{m_1 - 1}{m_1 - 1 + m_2 - 1} S_1^2 + \frac{m_2 - 1}{m_1 - 1 + m_2 - 1} S_2^2$

(利用自由度加权平均) $= \frac{m_1 - 1}{m_1 + m_2 - 2} S_1^2 + \frac{m_2 - 1}{m_1 + m_2 - 2} S_2^2$

证明: 当原假设成立时, 检验统计量

$$t_0 = \frac{\bar{y}_1 - \bar{y}_2}{S_w \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}} \sim t(m_1 + m_2 - 2)$$

独立同分布, $\bar{y}_1 \sim N(\mu_1, \frac{\sigma^2}{m_1})$ $\bar{y}_2 \sim N(\mu_2, \frac{\sigma^2}{m_2})$

$$\bar{y}_1 - \bar{y}_2 \sim N(\mu_1 - \mu_2, \frac{\sigma^2}{m_1} + \frac{\sigma^2}{m_2})$$

对 $\bar{y}_1 - \bar{y}_2$ 标准化, 则有 $\frac{\bar{y}_1 - \bar{y}_2}{\sigma \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}} \sim N(0, 1)$ (假设成立时, $\mu_1 = \mu_2$)
 利用到性质: $\text{Var}(ax) = a^2 \text{Var}(x)$

回顾: χ^2 分布: 若 $x_1, x_2, \dots, x_n$ 来自正态分布 $N(\mu, \sigma^2)$

其样本方差 $s^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2$, 则 $\frac{(m-1)s^2}{\sigma^2} \sim \chi^2(m-1)$
 $\left(\frac{\sum_{i=1}^m (x_i - \bar{x})^2}{\sigma^2} \right) \sim \chi^2(m-1)$

对于两样本, 有 $\frac{(m_1-1)s_1^2}{\sigma^2} \sim \chi^2(m_1-1)$, $\frac{(m_2-1)s_2}{\sigma^2} \sim \chi^2(m_2-1)$

由 χ^2 分布可加性, $\frac{(m_1-1)s_1^2 + (m_2-1)s_2^2}{\sigma^2} = \frac{(m_1+m_2-2)s_w^2}{\sigma^2} \sim \chi^2(m_1+m_2-2)$

t 分布定义: 若 $Z \sim N(0, 1)$, $W \sim \chi^2(n)$, 且 Z 与 W 相互独立

$$\text{则 } T = \frac{Z}{\sqrt{\frac{W}{n}}} \sim t(n)$$

$$\text{令 } Z = \frac{\bar{y}_1 - \bar{y}_2}{\sigma \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}}, \quad W = \frac{(m_1+m_2-2)s_w^2}{\sigma^2}, \quad \text{代入}$$

$$T = \frac{\frac{\bar{y}_1 - \bar{y}_2}{\sigma \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}}}{\sqrt{\frac{(m_1+m_2-2)s_w^2}{\sigma^2 (m_1+m_2-2)}}} = \frac{\frac{\bar{y}_1 - \bar{y}_2}{\cancel{\sigma} \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}}}{\frac{s_w}{\cancel{\sigma}}} = \frac{\bar{y}_1 - \bar{y}_2}{s_w \sqrt{\frac{1}{m_1} + \frac{1}{m_2}}} \sim T(m_1+m_2-2)$$

特别地, 当 $m_1 = m_2 = m$ 时, $t_0 = \frac{\bar{y}_1 - \bar{y}_2}{s_w \sqrt{\frac{2}{m}}} \sim t(2(m-1))$

两种检验方法：临界值法：查表， t 的绝对值不超过临界值，则接受
 p 值法：查表 $p > \alpha$ ，则接受原假设

均值模型：

$$y_{ij} = \mu_i + \varepsilon_{ij}, \quad i=1, 2, \dots, a, \quad j=1, 2, \dots, m \quad (a \text{ 个水平, 每个水平 } m \text{ 个值})$$

y_{ij} : 第 i 个水平下观测到的第 j 个响应变量 可理解为 a 组, 每组 m 个)

μ_i : 在第 i 个水平的均值 ε_{ij} : 随机误差

效应模型：

$$\mu_i = \mu + \alpha_i \quad (i=1, 2, \dots, a), \quad y_{ij} = \mu + \alpha_i + \varepsilon_{ij} \quad (i=1, 2, \dots, a, \quad j=1, 2, \dots, m)$$

μ : 总体均值 α_i : 第 i 个水平下的效应

通常, 有 $\sum_{i=1}^a \alpha_i = 0$, 为了避免模型冗余

eg: $\mu=10, \alpha_1=1, \alpha_2=2$ 与 $\mu=9, \alpha_1=2, \alpha_2=3$ 等效

为什么多个总体均值比较, 不能两两 t 检验?

会放大错误率。假设 3 次检验, 错误率为 0.05。两两 t 检验后, 错误率为

$$1 - (1 - 0.05)^3 \approx 0.14$$

平方和分解公式:

$$\sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{..})^2 = m \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{i.})^2$$

$\bar{y}_{..}$: 总均值, $\bar{y}_{i.}$: 第 i 组的均值

$$SS_T: \sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{..})^2 \text{ (总偏差平方和)}$$

$$SSA = m \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 \text{ (组间偏差平方和)}$$

计算方法: 每组均值与总均值的偏差平方和, 乘以样本量 m

$$SSE = \sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{i.})^2 \text{ (组内偏差平方和)}$$

$$SST = SSA + SSE$$

证明

$$f_2 = \sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{..})^2$$

$$= \sum_{i=1}^a \sum_{j=1}^m \left[(y_{ij} - \bar{y}_{i.}) + (\bar{y}_{i.} - \bar{y}_{..}) \right]^2$$

$$= \sum_{i=1}^a \sum_{j=1}^m \left[(y_{ij} - \bar{y}_{i.})^2 + 2(y_{ij} - \bar{y}_{i.})(\bar{y}_{i.} - \bar{y}_{..}) + (\bar{y}_{i.} - \bar{y}_{..})^2 \right]$$

$$= \underbrace{\sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{i.})^2}_{= SSE} + \underbrace{2 \sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{i.})(\bar{y}_{i.} - \bar{y}_{..})}_{= 0} + \underbrace{\sum_{i=1}^a \sum_{j=1}^m (\bar{y}_{i.} - \bar{y}_{..})^2}_{= m \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 = SSA}$$

$$= SSE$$

$$= 0$$

$$= m \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 = SSA$$

交叉项为0证明:

$$2 \sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot}) (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})$$

$$= 2 \sum_{i=1}^a (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \left[\sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot}) \right]$$

$$= 0$$

对于一个给定的 i , $\sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot}) = \sum_{j=1}^m y_{ij} - m \bar{y}_{i\cdot} = 0$

给定一组数据, SS_T 不会变, 构造 $\frac{SS_A}{SS_T}$, $\frac{SS_A}{SS_T}$ 越大, 不同组的影响越大, 假设不成立 (原假设为 $\mu_1 = \mu_2 = \dots = \mu_a$)

在单因子方差分析模型中, 使用的检验统计量为 $\frac{SS_A}{SS_E}$

$$1. \frac{SS_E}{\sigma^2} \sim \chi^2(n-a)$$

证明: 使用效应模型 $y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$

$$SS_E = \sum_{i=1}^a \sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot})^2$$

$$= \sum_{i=1}^a \sum_{j=1}^m \left((\mu + \alpha_i + \varepsilon_{ij}) - (\mu + \alpha_i + \frac{1}{m} \sum_{j=1}^m \varepsilon_{ij}) \right)^2$$

$$= \sum_{i=1}^a \sum_{j=1}^m (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2$$

ε_{ij} 独立同分布, 且 $\varepsilon_{ij} \sim N(0, \sigma^2)$

$$SS_E = \sum_{i=1}^a \sum_{j=1}^m (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2$$

将 $\varepsilon_{i1}, \varepsilon_{i2}, \dots, \varepsilon_{im}$ 看作独立同分布的样本, $\bar{\varepsilon}_{i\cdot}$ 看作样本均值

$$\therefore \sum_{j=1}^m \frac{(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2}{\sigma^2} \sim \chi^2(m-1), i=1, 2, \dots, a$$

由 χ^2 分布可知, $\frac{SS_E}{\sigma^2} \sim \chi^2(a(m-1))$ (a组, 每组自由度为 $m-1$)
 $= \chi^2(n-a)$ ($n=am$, 代表总数据量)

$$2. E(SS_A) = (a+1)\sigma^2 + m \sum_{i=1}^a \alpha_i^2$$

$$\begin{aligned} \text{证明: } SS_A &= m \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 \\ &= m \sum_{i=1}^a \left(\frac{1}{m} \sum_{j=1}^m (\mu + \alpha_i + \varepsilon_{ij}) - \frac{1}{n} \sum_{i=1}^a \sum_{j=1}^m (\mu + \alpha_i + \varepsilon_{ij}) \right)^2 \\ &= m \sum_{i=1}^a \left(\mu + \alpha_i + \bar{\varepsilon}_{i.} - (\mu + \bar{\varepsilon}_{..}) \right)^2 \\ &= m \sum_{i=1}^a (\alpha_i + \bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..})^2 \\ &= m \sum_{i=1}^a (\alpha_i^2 + (\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..})^2 + 2\alpha_i(\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..})) \\ &= m \sum_{i=1}^a \alpha_i^2 + m \sum_{i=1}^a (\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..})^2 + 2m \sum_{i=1}^a \alpha_i(\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..}) \end{aligned}$$

$\therefore \varepsilon_{ij}$ 独立同分布, $\varepsilon_{ij} \sim N(0, \sigma^2)$

$\therefore \bar{\varepsilon}_{i.} = \frac{1}{m} \sum_{j=1}^m \varepsilon_{ij} \sim N(0, \frac{\sigma^2}{m})$, $\bar{\varepsilon}_{..} \sim N(0, \frac{\sigma^2}{n})$

$$\therefore E(\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..}) = 0$$

$$\therefore E(SS_A) = m \sum_{i=1}^a \alpha_i^2 + m E\left(\sum_{i=1}^a (\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..})^2\right) + 0$$

期望为 0

Q: 既然 $E(\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..}) = 0$, $E((\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..})^2)$ 为何不为 0?

A: 一个统计量期望为 0, 其平方作为统计量, 期望不一定为 0, 而为其方差

$$\text{令 } E(X) = 0, Y = X^2$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = E(X^2) - 0 = E(X^2) = E(Y)$$

$$E(SS_A) = m \sum_{i=1}^a \alpha_i^2 + m \sum_{i=1}^a \text{Var}(\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..}) = m \sum_{i=1}^a \alpha_i^2 + m \cdot a \cdot \left(\frac{\sigma^2}{m} + \frac{\sigma^2}{am} \right) = m \sum_{i=1}^a \alpha_i^2 + (a+1)\sigma^2$$

$\therefore \bar{\varepsilon}_{..} = \frac{1}{a} \sum_{i=1}^a \bar{\varepsilon}_{i.}$, 且 $\bar{\varepsilon}_{i.}$ 之间相互独立, 且 $\bar{\varepsilon}_{i.} \sim N(0, \frac{\sigma^2}{m})$

$$\therefore \frac{1}{\sigma^2} \sum_{i=1}^a (\bar{\varepsilon}_{i.} - \bar{\varepsilon}_{..}) \sim \chi^2(a-1)$$

原假设成立时 $\alpha_1 = \alpha_2 = \dots = 0$

}, 有 $\frac{SSA}{\sigma^2} = \frac{m \sum_{i=1}^a (\bar{\epsilon}_{i.} - \bar{\epsilon}_{..})^2}{\sigma^2} \sim \chi^2(a-1)$

4. SSA 与 SS_E 独立, $SSA = m \sum_{i=1}^a (\alpha_i + \bar{\epsilon}_{i.} - \bar{\epsilon}_{..})^2$, 是 $\bar{\epsilon}_{1.}, \bar{\epsilon}_{2.}, \dots, \bar{\epsilon}_{a.}$ 的函数

$$SS_E = \sum_{i=1}^a \sum_{j=1}^m (\epsilon_{ij} - \bar{\epsilon}_{i.})^2$$

检验统计量

$$F_A = \frac{MS_A}{MS_E} = \frac{\frac{SSA}{a-1}}{\frac{SS_E}{n-a}} \sim F(a-1, n-a)$$

方差分析表

来源	平方和	自由度	均方	F 值	p 值
因子	SS_A	$a - 1$	$MS_A = \frac{SS_A}{a - 1}$	$F_A = \frac{MS_A}{MS_E}$	$p = P(F \geq F_A)$
误差	SS_E	$n - a$	$MS_E = \frac{SS_E}{n - a}$		
总和	SS_T	$n - 1$			

点估计: 假设每个数据 $y_{ij} \sim N(\mu + \alpha_i, \sigma^2)$

则需要估计的参数为 $(\mu, \alpha_1, \alpha_2, \dots, \alpha_a, \sigma^2)^T$, 共 $(a+2)$ 个